

JYOSTHNA LUNAVATH

Immunology & Cell Biology · Cell-Based Profiling (Imaging & Flow Cytometry) · Assay Development
London, UK (open to relocation; commutable to Oxford) · jyosthna.lunavath.23@ucl.ac.uk · +44 7751 219906
linkedin.com/in/jyosthna-lunavath-019796260

SUMMARY

Immunologist and cell-biology / assay-development scientist, final-year Ph.D. candidate at UCL (submitting Sep 2026). My master's research was a full immunology project in Prof. S. Murty Srinivasula's laboratory (Director, IISER), and my PhD built a multiparametric high-content imaging platform for CFTR. I profile cells by confocal imaging, culture primary human epithelial cells under sterile conditions, and run assays end to end with rigorous statistical QC, working with exceptional independence and across collaborative, cross-disciplinary teams. Keen to bring this immunology grounding and profiling expertise to translational research on immune-mediated inflammatory disease.

CORE COMPETENCIES

Immunology	Flow Cytometry	High-Content Imaging	High-Throughput Screening
Assay Development	Assay Optimisation & Validation	Z' / SSMD / QC Metrics	Dose-Response & EC50
Patch-Clamp Electrophysiology	Ion Channel Physiology & CFTR	Primary Human Cell Culture	iPSC & Organoids
Molecular Biology (qPCR / IVT)	Confocal & Live-Cell Imaging	Python · MATLAB · R	Translational Drug Discovery

RESEARCH EXPERIENCE

Doctoral Researcher, Cell-Based Assay Development & Cell Biology · University College London Sep 2023–Present
Division of Biosciences, Dept. of Neuroscience, Physiology & Pharmacology (Prof. Paola Vergani, primary supervisor; Dr. Cally Haynes)

- Designed, built, and validated a **multiparametric high-content imaging platform for CFTR**: one 96-well assay profiling protein biogenesis / subcellular localisation, ion-channel function, and intracellular pH together in live human epithelial cells, taken from concept to a validated screening tool.
- Ran **cell-based profiling by both imaging and flow cytometry**, developed functional fluorescence assays (halide-sensitive YFP), and set up statistical QC (Z', SSMD, signal-to-background).
- Built automated image-segmentation and analysis pipelines (Python/MATLAB/ImageJ) with UCL Computer Science & UCL Advance Research Computing, raising throughput ~50% and cutting analysis time ~40%; profiled compounds and validated hits with orthogonal assays (patch-clamp, Western blot, co-IP).
- Worked in **collaborative, cross-institution teams** (UCL, Institut Necker-Enfants Malades, Bristol, Erasmus MC), applying the platform to patient-derived cells with clinicians. Independently diagnosed and repaired a complex confocal fault; regarded by my PI and department as the 'world ImageXpress expert'.

Master's Research Project, Immunology & Molecular Biology · IISER Thiruvananthapuram, India Jul 2022–Apr 2023
Immunology laboratory of Prof. S. Murty Srinivasula (Director, IISER Thiruvananthapuram)

- Studied the Ubiquitin Proteasome System, E3 ligases **STUB1 and CARP2 in immune and cell-death (NF-κB) signalling**; used inflammation assays, co-immunoprecipitation, Western blotting, and molecular-biology techniques; characterised STUB1 variants and identified a CARP2-mediated regulatory axis. Designed and interpreted experiments independently.

Lipid Nanoparticle (LNP) Delivery: Formulation, Characterisation & Profiling · UCL 2023–Present
Depts. of Chemistry and NPP (Dr. Cally Haynes, Prof. Paola Vergani)

- Formulated and characterised surface-charged liposomal nanocarriers (DLS, zeta potential) and profiled delivery in HEK293 reporter cells; co-supervised the project with UCL Chemistry.

TECHNICAL SKILLS

Immunology	Master's research in Prof. S. Murty Srinivasula's immunology lab (IISER): E3 ligases STUB1 & CARP2 in immune / cell-death (NF-κB) signalling; inflammation assays; co-IP; Western blot; molecular biology
Assay Platforms	High-content imaging (HCI); high-throughput screening (HTS); halide-sensitive YFP; HTRF; 96/384-well formats; vesicle efflux assays; patch-clamp electrophysiology; flow cytometry
Assay Development & QC	Cell-based assay design, optimisation & validation; dose-response & EC50; Z'-factor, SSMD, signal-to-background; fluorescence spectral optimisation; ion channel physiology & CFTR
Cell Systems	Primary human bronchial & intestinal epithelial cells; iPSCs; organoid / 3D culture; HEK293; HeLa; sterile culture & sample handling (BSL-2)
Molecular Biology	qPCR / RT-qPCR; PCR; molecular cloning, bacterial culture & plasmid prep; transfection (lipofection, electroporation / nucleofection); in vitro transcription (IVT); gel electrophoresis; RNA / DNA extraction
Microscopy & Imaging	Molecular Devices ImageXpress confocal (expert), OperaPhenix Plus; confocal / widefield / live-cell / TIRF / DIC; custom fluorescence filter design; ImageJ/Fiji; CellProfiler; Cellpose; semi-automated image pipelines
Biochemistry & Analytical	Co-immunoprecipitation; Western blot; protein expression & purification; HPLC; FPLC; chromatography

Drug Discovery & Delivery	Compound profiling & hit validation; patient-derived models; molecular docking; ML-guided compound prioritisation; large-scale library analysis; LNP formulation & characterisation (DLS, zeta potential)
Software & Data	Python; MATLAB; R; Excel; GraphPad Prism; ImageJ/Fiji; ANOVA & statistical modelling; data visualisation; electronic lab notebook (ELN) & data integrity

EDUCATION

Ph.D., Pharmacology & Physiology · University College London 2023–Present (Ph.D. expected Sep 2026)
Cell-based assay development · cell & molecular biology · translational pharmacology · drug discovery

Integrated Master’s (BS-MS), Biological Sciences (Minor: Physics) · IISER Thiruvananthapuram, India 2023 · GPA 9.0 / 10
Immunology research project (Prof. S. Murty Srinivasula); IVT, RNA & molecular biology

PUBLICATIONS (IN PREPARATION)

- **Lunavath J., et al.** *High-Content fluorescence assay to simultaneously monitor CFTR biogenesis and ion-channel function in HEK293 cells.* (first author, in preparation)
- **Lunavath J., et al.** *High-Content Characterisation of CFTR-Mediated Bicarbonate Transport.* (first author, in preparation)
- **Lunavath J. (second author).** *GoSlo, a BK Channel Modulator, Enhances CFTR Activity.* (in preparation)

AWARDS & CERTIFICATIONS

- **SGP Rising Star Award**, Society of General Physiologists (2025, presented by Nobel laureate Prof. Roderick MacKinnon)
- **European CF Young Investigator Travel Award**, European Cystic Fibrosis Society (2026)

Certifications: Good Laboratory Practice (GLP) · Laboratory Animal Science & Handling (accredited) · GMP awareness · BSL-2 practice · electronic lab notebook (ELN) & data integrity

SELECTED CONFERENCE PRESENTATIONS

European Cystic Fibrosis Conference	Malta · Mar 2026
European CF Young Investigators’ Meeting	Paris, France · Feb 2026
UK Cystic Fibrosis Conference	London, UK · Oct 2025/2024/2023
Society of General Physiologists Annual Meeting	Boston, USA · Sep 2025
European Cystic Fibrosis Conference	Pisa, Italy · Mar 2025
Francis Crick Institute PhD Symposium	London, UK · Jul 2024
European Cystic Fibrosis Conference	Glasgow, UK · Jun 2024
Frontiers in Biology Conference	IISER Thiruvananthapuram · Mar 2023

LEADERSHIP & MENTORSHIP

- **Elected Faculty Representative**, UCL Faculty of Life Sciences (2025–Present). Promoted from NPP Departmental Representative (2024–2025); represent postgraduate researchers across one of UCL’s 11 faculties and engage senior university and sector leaders on education policy.
- **Teaching Assistant, In2Science Tutor & Research Mentor**, UCL (2023–Present). Taught 300+ students; supervised and trained ten students (five as full dissertation projects); manage all lab procurement and operations independently.

REFERENCES

- **Prof. Paola Vergani** · Primary PhD supervisor (PI), University College London
- **Prof. S. Murty Srinivasula** · Director, IISER Thiruvananthapuram (Master’s immunology research supervisor)
- **Dr. Naresh Kasoju** · Sree Chitra Tirunal Institute for Medical Sciences & Technology (SCTIMST), Thiruvananthapuram
- **Dr. Cally Haynes** · Secondary PhD supervisor, University College London
- **Dr. Guy Moss** · University College London

Full contact details available on request.

Collaborators:

Prof. David N. Sheppard - University of Bristol, UK
Prof. Isabelle Sermet-Gaudelus - Institut Necker-Enfants Malades, Paris, France
Dr. Marcel J. C. Bijvelds - Erasmus MC, Rotterdam, Netherland